

The Critical-Line Theorem

Arithmetic proof, quadratic-form consequences, and dictionary

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1. Statement and notation

This manuscript gives a complete deduction of the critical-line property from explicitly stated arithmetic premises. The cancellation estimate is retained as a premise, either through the singleton case of the prime-correlation axiom or through a short-interval error axiom. The analytic deductions are proved below. Sections 13-14 supply the dictionary and the exact dependency record.

Write

$$\Lambda(n) = \begin{cases} \log p, & n = p^k, \text{ } p \text{ prime, } k \geq 1, \\ 0, & \text{otherwise,} \end{cases} \quad \psi(x) = \sum_{n \leq x} \Lambda(n), \quad R(x) = \psi(x) - x. \quad (1)$$

Define the completed function

$$\xi(s) = \frac{1}{2} s(s-1) \pi^{-s/2} \Gamma(s/2) \zeta(s). \quad (2)$$

The target conclusion is

$$\boxed{\xi(\rho) = 0 \implies \Re \rho = \frac{1}{2}.} \quad (3)$$

The key reduction is the identity $\mathfrak{S}(\{0\}) = 1$. Consequently, the singleton case of the prime-correlation axiom already supplies $R(x) = O_\varepsilon(x^{1/2+\varepsilon})$. Sections 3-7 prove every subsequent implication. Section 8 proves an alternative route to the same error estimate.

2. Arithmetic premises

2.1. Axiom A: the Dirichlet series and Euler product

For $\Re s > 1$,

$$\zeta(s) = \sum_{n=1}^{\infty} n^{-s} = \prod_p (1 - p^{-s})^{-1}. \quad (4)$$

These fix the arithmetic function and its normalization. We use the usual arithmetic of the positive integers and the usual complex exponential. The series converge normally in smaller half-planes $\Re s \geq 1 + \delta$, $\delta > 0$.

For completeness, $0 \leq \Lambda(n) \leq \log n$, so

$$\sum_{n \geq 2} \Lambda(n) n^{-1-\delta} \leq \sum_{n \geq 2} (\log n) n^{-1-\delta} < \infty. \quad (5)$$

The logarithm of the product can therefore be differentiated term by term on such half-planes, giving

$$-\frac{\zeta'}{\zeta}(s) = \sum_{n \geq 1} \Lambda(n) n^{-s}, \quad \Re s > 1. \quad (6)$$

Indeed, $\log \zeta(s) = \sum_p \sum_{k \geq 1} p^{-ks}/k$, and differentiating produces $\sum_{p,k} (\log p) p^{-ks}$. The convergent logarithm also shows that the Euler product is nonzero there.

2.2. Axiom D: entire completion and functional equation

The function ξ in (2), initially defined for $\Re s > 1$, extends to an entire function and satisfies

$$\xi(s) = \xi(1-s). \quad (7)$$

This specifies the functional equation and analytic continuation. The present proof will explicitly handle the cancellation at $s = 1$; it will not declare the prefactor in (2) nonvanishing there.

2.3. Axiom E: the prime-correlation estimate

For an admissible finite set of shifts $H = \{h_1, \dots, h_k\}$, the stated prime-correlation premise is

$$\sum_{n \leq x} \prod_{j=1}^k \Lambda(n + h_j) = \mathfrak{S}(H)x + E_k(x; H), \quad (8)$$

where

$$\mathfrak{S}(H) = \prod_p \left(1 - \frac{\nu_p(H)}{p} \right) \left(1 - \frac{1}{p} \right)^{-k}, \quad (9)$$

and the error has the square-root bound $|E_k(x; H)| \leq x^{1/2+\varepsilon}$ for sufficiently large x , for each $\varepsilon > 0$, in the stated range of shifts. Only $k = 1$ and $H = \{0\}$ will be used. A version with an ε -dependent constant gives the identical conclusion.

The singleton $\{0\}$ occupies one residue class modulo each prime, hence is admissible. Its diameter is zero, so it satisfies the stated diameter restriction at every positive scale.

Logical status. Axiom E is a premise of the theorem. Its square-root error term is not derived from unique factorization. The proof below uses only its precisely identified singleton specialization.

3. The singleton theorem

Theorem 3.1. Axiom E implies that for every $\varepsilon > 0$ there is a finite constant B_ε with

$$|R(x)| \leq B_\varepsilon x^{1/2+\varepsilon} \quad (x \geq 1). \quad (10)$$

Proof. For $H = \{0\}$, $k = 1$ and $\nu_p(H) = 1$ for every prime. Each factor of (9) is exactly

$$\left(1 - \frac{1}{p}\right) \left(1 - \frac{1}{p}\right)^{-1} = 1.$$

Thus every finite partial product is 1, and $\mathfrak{S}(\{0\}) = 1$. Equation (8) becomes

$$\psi(x) = x + E_1(x; \{0\}).$$

Its error bound gives (10) for sufficiently large x . On the remaining compact interval, ψ has finitely many jumps and is bounded; hence so is $|R(x)|x^{-1/2-\varepsilon}$. Increasing the constant covers that interval. $\square$

Corollary 3.2. The mean-field law $\psi(x)/x \rightarrow 1$ follows from Axiom E.

Proof. Take $\varepsilon = 1/4$ in (10). Then

$$\left| \frac{\psi(x)}{x} - 1 \right| \leq B_{1/4} x^{-1/4} \rightarrow 0.$$

In particular $\psi(x) = O(x)$. $\square$

No multi-shift correlation, moment expansion, probability distribution, or large-deviation estimate has entered the proof.

4. The exact Mellin identity

Lemma 4.1. For $\Re s > 1$,

$$-\frac{\zeta'}{\zeta}(s) = \frac{s}{s-1} + s \int_1^\infty R(x) x^{-s-1} dx. \quad (11)$$

Proof. For finite X , partial summation of the step function ψ gives

$$\sum_{n \leq X} \Lambda(n) n^{-s} = \psi(X) X^{-s} + s \int_1^X \psi(x) x^{-s-1} dx. \quad (12)$$

There is no lower boundary contribution because $\Lambda(1) = 0$ and $\psi(1) = 0$. If $\sigma = \Re s > 1$, Corollary 3.2 gives

$$|\psi(X) X^{-s}| = O(X^{1-\sigma}) \rightarrow 0.$$

The integral is absolutely convergent. Letting $X \rightarrow \infty$ and using (6),

$$-\frac{\zeta'}{\zeta}(s) = s \int_1^\infty \psi(x)x^{-s-1} dx.$$

Substitute $\psi(x) = x + R(x)$. Since

$$s \int_1^\infty x^{-s} dx = \frac{s}{s-1},$$

we obtain (11). $\square$

5. Holomorphy throughout the right half-plane

Set

$$\mathcal{H} = \{s \in \mathbb{C} : \Re s > \tfrac{1}{2}\}, \quad J(s) = \int_1^\infty R(x)x^{-s-1} dx. \quad (13)$$

Lemma 5.1. The integral defining J converges absolutely at every point of $\mathcal{H}$, locally uniformly there, and defines a holomorphic function on all of $\mathcal{H}$.

Proof. Let $K \subset \mathcal{H}$ be compact and let

$$\delta = \min_{s \in K} (\Re s - \tfrac{1}{2}) > 0.$$

Choose $\varepsilon = \delta/2$ in (10). For $s \in K$ and $x \geq 1$,

$$|R(x)x^{-s-1}| \leq B_{\delta/2} x^{-1-\delta/2}. \quad (14)$$

The right side is integrable over $[1, \infty)$. It proves absolute convergence and uniform convergence on K . Each derivative in s of the integrand introduces a factor $-\log x$. For every fixed nonnegative integer j ,

$$\int_1^\infty x^{-1-\delta/2} (\log x)^j dx < \infty. \quad (15)$$

To see this directly, substitute $u = \log x$: the integral becomes $\int_0^\infty e^{-\delta u/2} u^j du$. Thus differentiation under the integral is valid on smaller neighborhoods of every point, and J is holomorphic. Since K was arbitrary, the conclusion holds throughout $\mathcal{H}$. $\square$

It follows that

$$M(s) = \frac{s}{s-1} + sJ(s) \quad (16)$$

is meromorphic on $\mathcal{H}$, holomorphic except at $s = 1$, where it has residue 1.

6. Meromorphic continuation and the pole at one

Axiom D provides the meromorphic continuation of ζ to $\mathcal{H}$ by

$$\zeta(s) = \frac{2\pi^{s/2}\xi(s)}{s(s-1)\Gamma(s/2)}. \quad (17)$$

Here $s \neq 0$ on $\mathcal{H}$, and $\Gamma(s/2)$ is holomorphic and nonzero there. Thus $s = 1$ is the only possible pole in $\mathcal{H}$. This continuation agrees with (4) on $\Re s > 1$.

Lemma 6.1. The pole of ζ at 1 is simple with residue 1, and

$$\xi(1) = \xi(0) = \frac{1}{2}. \quad (18)$$

Proof. For real $\sigma > 1$, comparison of the decreasing function $x^{-\sigma}$ with its integral gives

$$\frac{1}{\sigma-1} \leq \sum_{n \geq 1} n^{-\sigma} \leq 1 + \frac{1}{\sigma-1}.$$

Multiplication by $\sigma - 1$ shows

$$\lim_{\sigma \downarrow 1} (\sigma - 1)\zeta(\sigma) = 1.$$

The meromorphic function (17) has at most a simple pole there. The nonzero limit identifies its pole and residue. Substitution into (2), using $\Gamma(1/2) = \sqrt{\pi}$, gives $\xi(1) = 1/2$. Reflection (7) gives $\xi(0) = 1/2$. $\square$

Lemma 6.2. As meromorphic functions on $\mathcal{H}$,

$$-\frac{\zeta'}{\zeta} = M. \quad (19)$$

Proof. The function ζ is not identically zero because it is nonzero for $\Re s > 1$. Its logarithmic derivative is therefore meromorphic on $\mathcal{H}$. Equation (11) establishes equality with M on the nonempty open subset $\Re s > 1$. The identity theorem for meromorphic functions on the connected domain $\mathcal{H}$ extends that equality to the whole domain. $\square$

7. The critical-line theorem

Theorem 7.1. Under Axioms A and D and the single-shift case of E, every zero of ξ has real part $1/2$.

Proof. Suppose first that ζ has a zero $\rho \in \mathcal{H}$. Lemma 6.1 excludes $\rho = 1$. Let its multiplicity be $m \geq 1$. In a neighborhood of ρ , holomorphic factorization gives

$$\zeta(s) = (s - \rho)^m u(s), \quad u(\rho) \neq 0.$$

Consequently

$$-\frac{\zeta'}{\zeta}(s) = -\frac{m}{s-\rho} - \frac{u'(s)}{u(s)}. \quad (20)$$

This has a pole of nonzero residue $-m$ at ρ . But (19) identifies it with M , which is holomorphic there. This contradiction excludes every zero of ζ in $\mathcal{H}$.

For $s \in \mathcal{H} \setminus \{1\}$, all prefactors in (2) are finite and nonzero. Therefore ξ has no zero there. Lemma 6.1 separately gives $\xi(1) = 1/2$, so ξ has no zero anywhere with $\Re s > 1/2$.

If ξ had a zero ρ with $\Re \rho < 1/2$, the reflection law would give the zero $1 - \rho$, whose real part is greater than $1/2$. That has just been excluded. Hence every zero lies on $\Re s = 1/2$. Multiplicity was arbitrary and no height restriction was imposed. $\square$

The complete chain is

Axiom E at $H = \{0\}$ $\Downarrow$ $R(x) = O_\varepsilon(x^{1/2+\varepsilon}) \quad (\text{every } \varepsilon > 0)$ $\Downarrow$ $-\zeta'/\zeta - s/(s-1)$ holomorphic on $\Re s > 1/2$ $\Downarrow$ ξ has no zeros to the right of the critical line $\Downarrow$ reflection $\Re \rho = 1/2$ for every zero of ξ .	(21)
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8. Alternative arithmetic premise: short-interval error bounds

The same conclusion follows from Axioms A, C, and D. The required passage from short-interval estimates to a global error bound is proved here with every summand inside the specified range.

Axiom C states that for every $\varepsilon > 0$, for sufficiently large t , and for $t^\varepsilon \leq h \leq t$,

$$|\psi(t+h) - \psi(t) - h| \leq h^{1/2}t^\varepsilon. \quad (22)$$

Lemma 8.1. Axiom C implies (10).

Proof. Fix the desired exponent $\varepsilon > 0$, and put $e = \min\{\varepsilon, 1/4\}$, $\alpha = 1/2 + e$. Choose $X_0 \geq 2$ so that Axiom C with exponent e applies for $t \geq X_0$.

Since $e < 1$, the choice $h = t$ satisfies $t^e \leq h \leq t$. Therefore

$$|R(2t) - R(t)| \leq t^\alpha \quad (t \geq X_0). \quad (23)$$

For $x \geq X_0$, choose the integer $K \geq 0$ such that

$$y = 2^{-K}x \in [X_0, 2X_0).$$

The exact telescoping identity is

$$R(x) = R(y) + \sum_{j=0}^{K-1} (R(2^{j+1}y) - R(2^j y)). \quad (24)$$

Every base point $2^j y$ is at least X_0 , so (23) applies to every summand. If $B = \sup_{X_0 \leq u \leq 2X_0} |R(u)| < \infty$, then

$$\begin{aligned} |R(x)| &\leq B + y^\alpha \sum_{j=0}^{K-1} 2^{j\alpha} \\ &= B + \frac{x^\alpha - y^\alpha}{2^\alpha - 1} \leq B + \frac{x^\alpha}{2^\alpha - 1}. \end{aligned} \quad (25)$$

The empty sum for $K = 0$ obeys the same estimate. Since $e \leq \varepsilon$, this is $O_\varepsilon(x^{1/2+\varepsilon})$. The bounded interval $[1, X_0]$ is absorbed into the constant as in Theorem 3.1. $\square$

Theorem 8.2. Axioms A, C, and D imply the critical line.

Proof. Lemma 8.1 supplies precisely (10). Sections 4-7 then apply without change. $\square$

Thus either the singleton case of Axiom E or Axiom C supplies the global error estimate (10). The remaining analytic argument is the same in both cases.

9. Consequences for the odd Weil form

For a real odd test function f , supported in $[-a, a]$ and normalized by $\|f\|_2 = 1$, put

$$F(z) = \int_{-a}^a f(x) e^{izx} dx, \quad F^\sharp(z) = \overline{F(\bar{z})}, \quad S(z) = F(z)/i. \quad (26)$$

Then S is entire and real on the real axis. With $\tilde{f}(x) = f(-x)$, let $g = f * \tilde{f}$. Its Fourier transform is $FF^\sharp = S^2$. The odd Weil form in the supplied normalization is

$$\begin{aligned} Q(f) &= \frac{1}{2\pi} \int_{\mathbb{R}} |F(t)|^2 [\Re \psi_\Gamma \left(\frac{1}{4} + \frac{it}{2} \right) - \log \pi] dt \\ &\quad - 2 \left(\int_{-a}^a f(x) \sinh(x/2) dx \right)^2 - 2 \sum_{n \geq 2} \frac{\Lambda(n)}{\sqrt{n}} g(\log n), \end{aligned} \quad (27)$$

where $\psi_\Gamma = \Gamma'/\Gamma$ is the digamma function. It is distinct from the arithmetic function $\psi(x)$ in (1).

The explicit-formula premise of the supplied manuscripts is

$$Q(f) = \sum_{\rho} F(\gamma_\rho - id_\rho) \overline{F(\gamma_\rho + id_\rho)}, \quad d_\rho = \Re \rho - \frac{1}{2}, \quad (28)$$

with the specified convergence and justified extension to the form domain. Under the critical-line conclusion already proved, $d_\rho = 0$, and therefore

$$Q(f) = \sum_{\rho} |F(\gamma_{\rho})|^2 \geq 0. \quad (29)$$

The sum includes the multiplicities of the zeros. For a finite symmetry orbit $\mathcal{O}$, define its contribution by summing the terms of (28) over that orbit. Each zero is counted once, with its multiplicity. Combining positive and negative ordinates accounts for a factor two inside the corresponding orbit; it does not introduce an additional factor two into a sum already taken over all zeros.

Define

$$\lambda(a) = \inf\{Q(f) : f \text{ real and odd, } \|f\|_2 = 1, \text{ supp } f \subset [-a, a]\}. \quad (30)$$

Equation (29) proves $\lambda(a) \geq 0$. The strict inequality $\lambda(a) > 0$ requires further justification. In particular, repeated zeta zeros do not impose corresponding orders of vanishing on F : a sum weighted by zero multiplicities must not be substituted for a count of distinct zeros of F .

The following independent differential theorem proves strict positivity under explicitly stated quadratic-form premises. Those premises are not needed in Sections 3-8.

10. Strict positivity from a relative derivative bound

Let $\lambda : (0, \infty) \rightarrow \mathbb{R}$ be locally absolutely continuous. Assume that there are $a_Z > 0$ and $\lambda_Z > 0$ with

$$\lambda(a) > 0 \quad (0 < a \leq a_Z), \quad \lambda(a_Z) \geq \lambda_Z. \quad (31)$$

Let $E(a) \geq 0$ be measurable, and suppose

$$\lambda'(a) = -E(a) \quad \text{a.e.} \quad (32)$$

For the boundary-amplitude normalization in the supplied manuscripts, $E(a) = 2|A_0(a)|^2$. Finally, assume a nonnegative function

$$C \in L^1_{\text{loc}}([a_Z, \infty))$$

satisfies $E(a) \leq C(a)\lambda(a)$ at almost every a where $\lambda(a) > 0$. Local integrability is required on the entire indicated range, including across any prospective first zero of λ .

Theorem 10.1. Under these premises,

$$\lambda(a) \geq \lambda_Z \exp \left(- \int_{a_Z}^a C(u) du \right) > 0 \quad (a \geq a_Z). \quad (33)$$

Proof. If a nonpositive value exists, continuity gives a first zero $a_d > a_Z$, with positivity on $[a_Z, a_d)$. On every compact subinterval of that range define

$$U(a) = \lambda(a) \exp \left(\int_{a_Z}^a C(u) du \right).$$

The product is absolutely continuous, and

$$U'(a) = e^{\int_{a_Z}^a C} (\lambda'(a) + C(a)\lambda(a)) \geq 0 \quad \text{a.e.}$$

It follows that $\lambda(a) \geq \lambda_Z e^{-\int_{a_Z}^a C}$. Since C is integrable through a_d , passage to the limit by continuity gives $\lambda(a_d) > 0$, a contradiction. The same integration then gives (33) at every later finite support parameter. $\square$

The conclusion covers every a , although (32) and its comparison need hold only almost everywhere. Absolute continuity is what justifies this passage.

If $\lambda > 0$ everywhere, the exact relative derivative

$$C_0(a) = -\frac{\lambda'(a)}{\lambda(a)}$$

belongs to L^1 on every compact support interval: λ has a positive minimum there and $\lambda' \in L^1$. Integration of $(\log \lambda)' = -C_0$ gives

$$\lambda(a) = \lambda(a_Z) e^{-\int_{a_Z}^a C_0}, \quad \lambda(a) e^{\int_{a_Z}^a C_0} = \lambda(a_Z). \quad (34)$$

This identity characterizes the exact relative derivative after strict positivity has been established; it is not an independent way to establish that positivity.

11. Signed spectral contributions

Suppose real symmetry-orbit contributions $w_{\mathcal{O}}(a)$ are absolutely summable and satisfy $\sum_{\mathcal{O}} w_{\mathcal{O}}(a) = \lambda(a)$. Define

$$N(a) = \sum_{\mathcal{O}} \max\{-w_{\mathcal{O}}(a), 0\}, \quad W_a(T) = \sum_{h_{\mathcal{O}} > T} w_{\mathcal{O}}(a). \quad (35)$$

The sum of the positive contributions equals $\lambda(a) + N(a)$. Therefore $W_a(T) \leq \lambda(a) + N(a)$.

Theorem 11.1. Retain (31)-(32). At almost every a where $\lambda(a) > 0$, suppose

$$N(a) \leq b(a)\lambda(a), \quad W_a(T_0(a)) \geq \eta(a) \frac{E(a)}{\pi T_0(a)} - \epsilon(a)\lambda(a), \quad (36)$$

where $T_0 > 0$, $0 < \eta \leq 1$, and $b, \epsilon \geq 0$ are measurable. If

$$K(a) = \frac{\pi T_0(a)}{\eta(a)} (1 + b(a) + \epsilon(a)) \in L_{\text{loc}}^1([a_Z, \infty)), \quad (37)$$

then $\lambda(a) > 0$ at every support parameter.

Proof. Equations (35)-(36) imply

$$\eta \frac{E}{\pi T_0} - \epsilon \lambda \leq W_a(T_0) \leq \lambda + N \leq (1 + b)\lambda.$$

Thus $E \leq K\lambda$, and Theorem 10.1 applies. $\square$

In the nonnegative case, $N = 0$. A lower tail estimate at $T_0 = c_{\text{on}} T^*(a)$, where $T^*(a) = 2\pi e^{2a}$, with fixed $\eta > 0$ and $\epsilon = 0$, yields

$$\lambda(a) \geq \lambda_Z \exp \left[-\frac{\pi c_{\text{on}}}{2\eta} (T^*(a) - T^*(a_Z)) \right]. \quad (38)$$

The tail estimate and the integrability in (37) remain premises of this theorem. A fixed-parameter tail asymptotic alone does not supply them.

12. The local mirror-pair expansion

Use the convention $F = iS$ of (26). For a real ordinate γ and a real displacement m , put

$$P_\gamma(m) = 2\Re \left[F(\gamma - im) \overline{F(\gamma + im)} \right].$$

Since $\overline{S(\bar{z})} = S(z)$, the product inside the real part is $S(\gamma - im)^2$. Taylor expansion gives

$$P_\gamma(m) = 2S(\gamma)^2 - 2m^2 (S'(\gamma)^2 + S(\gamma)S''(\gamma)) + O_a(m^4). \quad (39)$$

For normalized f , differentiation under its compactly supported Fourier integral gives

$$|F^{(j)}(\gamma + iy)| \leq \sqrt{2a} a^j e^{a|y|}.$$

On $|m| \leq 1/a$, this bounds the fourth derivative of $S(\gamma - im)^2$ uniformly in real γ and normalized f , with a constant depending on a . The real part is even in m , so Taylor's theorem gives the stated fourth-order remainder.

For the alternative convention $S_0(z) = \int_0^a f(x) \sin(zx) dx$, one has $F = 2iS_0$, and (39) becomes

$$P_\gamma(m) = 2|F(\gamma)|^2 - 2m^2 (|F'(\gamma)|^2 + 4S_0(\gamma)S_0''(\gamma)) + O_a(m^4). \quad (40)$$

This local calculation determines the second-order contribution of a fixed mirror pair. It does not establish a sign for the complete quadratic form by itself.

13. Dictionary

This dictionary is compiled from the definitions in the supplied manuscripts. It fixes both the mathematical objects and the translation of earlier terminology. Throughout Sections 1-12, the mathematical definitions are used directly.

13.1. Arithmetic and analytic notation

Symbol	Definition
$\Lambda(n)$	Von Mangoldt function: $\log p$ at $n = p^k$, zero otherwise.
$\psi(x)$	Chebyshev function $\sum_{n \leq x} \Lambda(n)$.
$\psi_\Gamma(z)$	Digamma function $\Gamma'(z)/\Gamma(z)$; distinct from $\psi(x)$.
$R(x)$	Arithmetic error $\psi(x) - x$.
$\zeta(s)$	Dirichlet series $\sum n^{-s}$ on $\Re s > 1$, with specified meromorphic continuation.
$\xi(s)$	Completed function in (2).
$\rho = \frac{1}{2} + d_\rho + i\gamma_\rho$	A zero, its signed horizontal displacement, and its ordinate.
m_ρ	Absolute displacement $ d_\rho $; separate from zero multiplicity.
$\mathfrak{S}(H)$	Singular series in (9), with $\nu_p(H)$ the number of represented residue classes.
a	Half-width of the support interval $[-a, a]$.
$F, F^\sharp, S$	Fourier transform, conjugate-reflected entire function, and $S = F/i$, as in (26).
$Q(f)$	Odd Weil quadratic form in (27).
$\lambda(a)$	Infimum of $Q(f)$ over normalized real odd test functions with support in $[-a, a]$.
$A_0(a)$	Boundary-amplitude coefficient of an attained minimizer, in the supplied logarithmic edge normalization.
$E(a)$	$2 A_0(a) ^2$, related to $-\lambda'(a)$ when the derivative identity is available.
$T^*(a)$	Reference ordinate scale $2\pi e^{2a}$; it does not impose exact spectral truncation.
$N(a)$	Total negative symmetry-orbit contribution in (35); not a zero-counting function.
C, K	Nonnegative locally integrable majorants of E/λ on the positive range.

13.2. Earlier terminology translated into number theory

Earlier term	Exact mathematical interpretation
Prime sheet	Logarithmic coordinate $u = \log n$ for arithmetic terms.
Charge at a prime power	Weight $\Lambda(n)n^{-1/2}$ in the explicit formula.
Zero sheet	The zero parameters (d_ρ, γ_ρ) .
Quantum	A zero of ξ , counted with its multiplicity.
Mass	Absolute horizontal displacement $ \Re \rho - 1/2 $.
Line Postulate	Every zero of ξ lies on $\Re s = 1/2$.
Observer and age	The test-function space on $[-a, a]$ and its half-width a .
Field and ground field	An admissible normalized test function and a minimizer attaining $\lambda(a)$, respectively.
Reserve or floor	The scalar $\lambda(a)$, the bottom of the normalized quadratic form.
Alive; threshold; negative	Respectively $\lambda(a) > 0$, $\lambda(a) = 0$, and $\lambda(a) < 0$.
Leakage	The derivative quantity $E(a) = -\lambda'(a)$, under the stated boundary identity.
Proportional leakage	The estimate $E(a) \leq C(a)\lambda(a)$, with C locally integrable through every finite support interval.

Earlier term	Exact mathematical interpretation
Packet	A finite orbit of zeros under reflection and conjugation, with its summed contribution to (28).
Born rule	Nonnegative normalized orbit contributions summing to one; normalization requires $\lambda(a) > 0$.
Horizon	The reference height $T^*(a) = 2\pi e^{2a}$.
Immortality	Strict positivity $\lambda(a) > 0$ for every finite $a > 0$.
Renormalized floor	The integrating-factor expression $\lambda(a)e^{\int C}$.
Random Walk Law	The deterministic short-interval bound (22), stated with its quantifiers.
Sieve Independence Law	The explicit prime-correlation estimate (8), including its error term.

The last two entries denote quantitative arithmetic premises. Their informal names supply no additional proof of those premises.

14. Exact dependency record

Result	Premises consumed	Conclusion
Theorem 3.1	Axiom E only at $H = \{0\}$.	Global error bound (10).
Lemma 8.1	Axiom C.	The same global error bound (10).
Sections 4-7	Axiom A, Axiom D, and (10).	Every zero of ξ has real part $1/2$.
Equation (29)	The critical-line conclusion and the explicit formula (28).	Nonnegativity of the odd Weil form.
Theorem 10.1	Positive initial range, absolute continuity, (32), and an integrable relative bound.	Strict positivity at every support parameter.
Theorem 11.1	The differential premises and the signed estimates (36)-(37).	An integrable relative bound, hence strict positivity.
Equation (34)	Strict positivity, absolute continuity, and (32).	Exact logarithmic-derivative representation.

The critical-line theorem is therefore

$$\boxed{\text{Axioms A and D} + (\text{E at } H = \{0\} \text{ or C}) \implies \Re \rho = \frac{1}{2} \text{ for all zeros of } \xi.} \quad (41)$$

All intermediate analytic steps in this implication have been supplied. The singleton prime-correlation estimate and the short-interval bound remain declared arithmetic premises. The quadratic-form consequences retain exactly the additional premises shown in the table; none is silently inferred from a zero-counting formula with multiplicities.

Source record

The argument uses the definitions and axioms in the supplied *The Line from the Random Walk of the Primes* (Manuscript II), the first-threshold and signed-form arguments in *The Immortality of Observers*, and the abbreviated formulation in *The Line from Proportional Leakage*. The present dictionary records their mathematical translations. The proof in Sections 3-8 is the earlier expanded proof, retained here with number-theoretic terminology throughout.